

Lab-17.2

Course : AI Assisted Coding

Topic: AI for Data Processing: Data Cleaning and Preprocessing Scripts

Name : Mohammed Sabir

Hall Ticket No : 2303A51506

Batch : 22

Task-01

Final Optimal Prompt:

Generate a Python script using Pandas to clean and preprocess a student academic dataset. Handle missing values in marks, attendance, and department, remove duplicate records based on student ID, standardize department names, and encode categorical variables like gender and department.

Code Screenshot :

```

lab19.py lab19.py\...
1  import pandas as pd
2
3  # Load dataset
4  # Replace 'students.csv' with your file name
5  df = pd.read_csv('C:\\Users\\sakir\\OneDrive\\Desktop\\Btech\\3 yr 2sem\\Ai-Assistant\\dataset19\\student.csv')
6  print("Original Data:")
7  print(df.head())
8  # -----
9  # 1. Handle Missing Values
10 # -----
11 # Fill missing marks with mean
12 df['marks'] = df['marks'].fillna(df['marks'].mean())
13 # Fill missing attendance with median
14 df['attendance'] = df['attendance'].fillna(df['attendance'].median())
15 # Fill missing department with mode
16 df['department'] = df['department'].fillna(df['department'].mode()[0])
17 # -----
18 # 2. Remove Duplicate Records
19 # -----
20 # Remove duplicates based on student_id
21 df = df.drop_duplicates(subset='student_id')
22 # -----
23 # 3. Standardize Department Names
24 # -----
25 # Convert to lowercase and strip spaces
26 df['department'] = df['department'].str.lower().str.strip()
27 # Replace variations with standard names
28 df['department'] = df['department'].replace({
29     'cse': 'computer science',
30     'cs': 'computer science',
31     'comp sci': 'computer science',
32     'ece': 'electronics',
33     'eee': 'electrical',
34     'mech': 'mechanical'
35 })
36 # -----
37 # 4. Encode Categorical Variables
38 # -----
39 # Encode gender (Label Encoding)
40 df['gender'] = df['gender'].str.lower().map({
41     'male': 0,
42     'female': 1,
43     'other': 2
44 })
45 # One-hot encode department
46 df = pd.get_dummies(df, columns=['department'])
47 # -----
48 # 5. Final Output
49 # -----
50 print("\nCleaned Data:")
51 print(df.head())
52 # Save cleaned dataset
53 df.to_csv('C:\\Users\\sakir\\OneDrive\\Desktop\\Btech\\3 yr 2sem\\Ai-Assistant\\dataset19\\cleaned_students.csv', index=False)
54 print("\nData cleaning completed and saved as 'cleaned_students.csv'")

```

Output Screenshot:

```
py-2025.18.0-win32-x64\bundled\libs\debugpy\launcher" "61123" "--" "c:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19\lab19.py"
Original Data:
  student_id  name  gender  marks  attendance  department
0         101   Amit   male    85.0         90.0         CSE
1         102   Sara  female   78.0         85.0         cs
2         103  Rahul   male    NaN         88.0    Comp Sci
3         104  Anita  female   92.0         NaN         ECE
4         105   John   male    67.0         75.0         eee

Cleaned Data:
  student_id  name  gender  marks  attendance  department_computer  science  department_electrical  department_electronics  department_mechanical
0         101   Amit     0    85.000000         90.0              True              False              False              False
1         102   Sara     1    78.000000         85.0              True              False              False              False
2         103  Rahul     0    80.846154         88.0              True              False              False              False
3         104  Anita     1    92.000000         88.0              False              False              True              False
4         105   John     0    67.000000         75.0              False              True              False              False
3         104  Anita  female   92.0         NaN         ECE
4         105   John   male    67.0         75.0         eee

Cleaned Data:
  student_id  name  gender  marks  attendance  department_computer  science  department_electrical  department_electronics  department_mechanical
0         101   Amit     0    85.000000         90.0              True              False              False              False
1         102   Sara     1    78.000000         85.0              True              False              False              False
2         103  Rahul     0    80.846154         88.0              True              False              False              False
3         104  Anita     1    92.000000         88.0              False              False              True              False
4         105   John     0    67.000000         75.0              False              True              False              False

Data cleaning completed and saved as 'cleaned_students.csv'
PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19>
```

Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

The student dataset was successfully cleaned and preprocessed to improve its quality and usability. Missing values in marks, attendance, and department were handled using appropriate statistical methods. Duplicate records were removed based on student ID, ensuring data uniqueness. Department names were standardized for consistency, and categorical variables like gender and department were encoded for analysis. These steps enhanced data reliability, reduced inconsistencies, and transformed the dataset into a structured format suitable for further analysis, visualization, and machine learning applications.

Task-02

Final Optimal Prompt:

Generate a Python script to preprocess financial transaction data. Convert date columns to datetime format, create new features like month and year, normalize the transaction amount, and detect and handle extreme values.

Code Screenshot :

```

61 import pandas as pd
62 import numpy as np
63 from sklearn.preprocessing import MinMaxScaler
64 # Load dataset
65 # Replace 'transactions.csv' with your file
66 df = pd.read_csv('C:\\Users\\sakir\\OneDrive\\Desktop\\Btech\\3 yr 2sem\\Ai-Assistant\\dataset19\\transactions.csv')
67 print("Original Data:")
68 print(df.head())
69 # -----
70 # 1. Convert Date Column
71 # -----
72 # Convert transaction_date to datetime
73 df['transaction_date'] = pd.to_datetime(df['transaction_date'], errors='coerce')
74 # -----
75 # 2. Feature Engineering
76 # -----
77 # Extract year, month, day
78 df['year'] = df['transaction_date'].dt.year
79 df['month'] = df['transaction_date'].dt.month
80 df['day'] = df['transaction_date'].dt.day
81 # Optional: weekday
82 df['weekday'] = df['transaction_date'].dt.day_name()
83 # -----
84 # 3. Normalize Transaction Amount
85 # -----
86 scaler = MinMaxScaler()
87 df['amount_normalized'] = scaler.fit_transform(df[['amount']])
88 # -----
89 # 4. Detect and Handle Extreme Values (Outliers)
90 # -----
91 # Using IQR method
92 Q1 = df['amount'].quantile(0.25)
93 Q3 = df['amount'].quantile(0.75)
94 IQR = Q3 - Q1
95 lower_bound = Q1 - 1.5 * IQR
96 upper_bound = Q3 + 1.5 * IQR
97 # Detect outliers
98 outliers = df[(df['amount'] < lower_bound) | (df['amount'] > upper_bound)]
99 print("\nOutliers detected:")
100 print(outliers)
101 # Option 1: Cap outliers (Winsorization)
102 df['amount_capped'] = np.where(df['amount'] > upper_bound, upper_bound,
103                               np.where(df['amount'] < lower_bound, lower_bound, df['amount']))
104 # Option 2 (alternative): Remove outliers
105 # df = df[(df['amount'] >= lower_bound) & (df['amount'] <= upper_bound)]
106 # -----
107 # 5. Handle Missing Values (Optional)
108 # -----
109 # Fill missing amounts with median
110 df['amount'] = df['amount'].fillna(df['amount'].median())
111 # Drop rows where date conversion failed
112 df = df.dropna(subset=['transaction_date'])
113 # -----
114 # Final Output
115 # -----
116 print("\nProcessed Data:")
117 print(df.head())
118 # Save cleaned dataset
119 df.to_csv('processed_transactions.csv', index=False)
120 print("\nData preprocessing completed and saved as 'processed_transactions.csv'")

```

Output Screenshot:

```

PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\AI-Assistant\dataset19> c:: cd 'c:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\AI-Assistant\dataset19'; & 'c:\python314\python.exe' 'c:\Users\sakir\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '49976' '--' 'c:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\AI-Assistant\dataset19\lab19.py'
Original Data:
transaction_id customer_id transaction_date amount payment_method location
0 T001 C101 2024-01-15 2500.0 UPI Hyderabad
1 T002 C102 2024-01-16 1800.0 Card Bangalore
2 T003 C103 2024-01-17 NaN Cash Chennai
3 T004 C104 2024-02-01 50000.0 UPI Mumbai
4 T005 C105 2024-02-05 1200.0 Card Delhi

Outliers detected:
transaction_id customer_id transaction_date amount payment_method location year month day weekday amount_normalized
3 T004 C104 2024-02-01 50000.0 UPI Mumbai 2024.0 2.0 1.0 Thursday 0.071362
6 T007 C106 2024-02-12 700000.0 Cash Kolkata 2024.0 2.0 12.0 Monday 1.000000
11 T012 C111 2024-03-07 99999.0 Cash Mumbai 2024.0 3.0 7.0 Thursday 0.142794
16 T017 C116 2024-03-20 250000.0 Card Bangalore 2024.0 3.0 20.0 Wednesday 0.357097

Processed Data:
transaction_id customer_id transaction_date amount payment_method location year month day weekday amount_normalized amount_capped
0 T001 C101 2024-01-15 2500.0 UPI Hyderabad 2024.0 1.0 15.0 Monday 0.003500 2500.0
1 T002 C102 2024-01-16 1800.0 Card Bangalore 2024.0 1.0 16.0 Tuesday 0.002500 1800.0
2 T003 C103 2024-01-17 1600.0 Cash Chennai 2024.0 1.0 17.0 Wednesday NaN NaN
3 T004 C104 2024-02-01 50000.0 UPI Mumbai 2024.0 2.0 1.0 Thursday 0.071362 9025.0
4 T005 C105 2024-02-05 1200.0 Card Delhi 2024.0 2.0 5.0 Monday 0.001643 1200.0

Data preprocessing completed and saved as 'processed_transactions.csv'
PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\AI-Assistant\dataset19>

```

Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

The dataset was successfully preprocessed by converting dates, extracting time-based features, and normalizing transaction amounts. Outliers were identified using the IQR method and handled through capping. Missing values were treated appropriately, resulting in a cleaner, structured dataset suitable for analysis and machine learning tasks.

Task-03

Final Optimal Prompt:

Generate a Python script to clean and preprocess environmental sensor data. Handle missing values, scale numerical features, encode categorical fields like sensor status, and split the dataset into training and testing sets.

Code Screenshot :

```
126     ***
127     import pandas as pd
128     from sklearn.preprocessing import StandardScaler, LabelEncoder
129     from sklearn.model_selection import train_test_split
130     # Load dataset
131     # Replace 'sensor_data.csv' with your file
132     df = pd.read_csv('C:\\Users\\sakir\\OneDrive\\Desktop\\Btech\\3 yr 2sem\\Ai-Assistant\\dataset19\\sensor_data.csv')
133     print("Original Data:")
134     print(df.head())
135     # -----
136     # 1. Handle Missing Values
137     # -----
138     # Fill numerical columns with mean
139     num_cols = df.select_dtypes(include=['float64', 'int64']).columns
140     df[num_cols] = df[num_cols].fillna(df[num_cols].mean())
141     # Fill categorical columns with mode
142     cat_cols = df.select_dtypes(include=['object']).columns
143     for col in cat_cols:
144         df[col] = df[col].fillna(df[col].mode()[0])
145     # -----
146     # 2. Scale Numerical Features
147     # -----
148     scaler = StandardScaler()
149     df[num_cols] = scaler.fit_transform(df[num_cols])
150     # -----
151     # 3. Encode Categorical Variables
152     # -----
153     # Example: encoding sensor_status
154     if 'sensor_status' in df.columns:
155         encoder = LabelEncoder()
156         df['sensor_status'] = encoder.fit_transform(df['sensor_status'])
157     # -----
158     # 4. Split Dataset
159     # -----
160     # Assume target column is 'target' (change if needed)
161     X = df.drop('target', axis=1)
162     y = df['target']
163     # Train-test split (80% train, 20% test)
164     X_train, X_test, y_train, y_test = train_test_split(
165         X, y, test_size=0.2, random_state=42
166     )
167     # -----
168     # Final Output
169     # -----
170     print("\nTraining set shape:", X_train.shape)
171     print("Testing set shape:", X_test.shape)
172     # Save processed datasets
173     X_train.to_csv('X_train.csv', index=False)
174     X_test.to_csv('X_test.csv', index=False)
175     y_train.to_csv('y_train.csv', index=False)
176     y_test.to_csv('y_test.csv', index=False)
177     print("\nData preprocessing completed and datasets saved.")
```

Output Screenshot:

```
PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19> c;; cd 'c:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19'; & 'c:\python314\python.exe' 'c:\Users\sakir\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '53675' '--' 'c:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19\lab19.py'
● Original Data:
  sensor_id temperature humidity pressure sensor_status location target
0      S001         25.5      60.0      1012.0         active   Room1         1
1      S002         27.2      65.0      1010.0        inactive   Room2         0
2      S003          NaN      58.0      1008.0         active   Room1         1
3      S004         30.1      NaN      1015.0        faulty    Room3         0
4      S005         22.8      55.0      1005.0         active   Room2         1
c:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19\lab19.py:142: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.
  cat_cols = df.select_dtypes(include=['object']).columns

Training set shape: (12, 6)
Testing set shape: (3, 6)

Data preprocessing completed and datasets saved.
PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19> |
```

Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

The environmental sensor dataset was effectively cleaned by handling missing values and standardizing numerical features through scaling. Categorical data, such as sensor status, was encoded for model compatibility. The dataset was then split into training and testing sets, ensuring proper evaluation of models. Overall, the preprocessing improved data quality, consistency, and readiness for further analysis or machine learning applications.

Task-04

Final Optimal Prompt:

Generate a Python script to clean smart city traffic data. Standardize road and location names, fill missing traffic density values, remove duplicate records, normalize speed and vehicle count, and provide a summary comparing data quality before and after cleaning.

Code Screenshot :

```

184 import pandas as pd
185 import numpy as np
186 from sklearn.preprocessing import MinMaxScaler
187 # Load dataset
188 # Replace 'traffic_data.csv' with your file
189 df = pd.read_csv(r"C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19\traffic_data.csv")
190 # -----
191 # Function to check data quality
192 # -----
193 def data_quality_summary(data, label):
194     print(f"\n--- {label} Data Quality Summary ---")
195     print("Shape:", data.shape)
196     print("\nMissing Values:\n", data.isnull().sum())
197     print("\nDuplicate Rows:", data.duplicated().sum())
198 # Initial summary
199 data_quality_summary(df, "Before Cleaning")
200 # -----
201 # 1. Standardize Road & Location Names
202 # -----
203 for col in ['road_name', 'location']:
204     if col in df.columns:
205         df[col] = df[col].str.lower().str.strip()
206 # Optional replacements (example)
207 df['road_name'] = df['road_name'].replace({
208     'mg rd': 'mahatma gandhi road',
209     'nh 44': 'national highway 44'
210 })
211 # -----
212 # 2. Handle Missing Values
213 # -----
214 # Fill traffic_density with median
215 if 'traffic_density' in df.columns:
216     df['traffic_density'] = df['traffic_density'].fillna(df['traffic_density'].median())
217 # -----
218 # 3. Remove Duplicates
219 # -----
220 df = df.drop_duplicates()
221 # -----
222 # 4. Normalize Numerical Features
223 # -----
224 scaler = MinMaxScaler()
225 num_cols = []
226 for col in ['speed', 'vehicle_count']:
227     if col in df.columns:
228         num_cols.append(col)
229 df[num_cols] = scaler.fit_transform(df[num_cols])
230 # -----
231 # 5. Final Summary
232 # -----
233 data_quality_summary(df, "After Cleaning")
234 # Save cleaned dataset
235 df.to_csv('cleaned_traffic_data.csv', index=False)
236 print("\nData cleaning completed and saved as 'cleaned_traffic_data.csv'")

```

Output Screenshot:


```

PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19> c:: cd 'c:\Users\sakir\OneDrive\
n.exe' 'c:\Users\sakir\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher
sistant\dataset19\lab19.py'

--- Before Cleaning Data Quality Summary ---
Shape: (15, 6)

Missing Values:
  road_name      0
  location       0
  traffic_density 2
  speed          1
  vehicle_count  1
  time           0
dtype: int64

Duplicate Rows: 1

--- After Cleaning Data Quality Summary ---
Shape: (14, 6)

Missing Values:
  road_name      0
  location       0
  traffic_density 0
  speed          1
  vehicle_count  1
  time           0
dtype: int64

Duplicate Rows: 0

Data cleaning completed and saved as 'cleaned_traffic_data.csv'
PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19> 

```

Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

The smart city traffic dataset showed significant improvement after preprocessing. Initially, it contained missing traffic density values, inconsistent road and location names, and duplicate records. After cleaning, missing values were filled, duplicates were removed, and text fields were standardized for consistency. Numerical features like speed and vehicle count were normalized, improving comparability across records. The data quality summary highlighted reduced errors and better structure, making the dataset more reliable and suitable for analysis, visualization, and intelligent traffic management system development.

Tack-05

Final Optimal Prompt:

Generate a Python script to clean and preprocess social media text data. Remove duplicates and null entries, clean text (remove URLs and special characters), convert to lowercase,

tokenize text, remove stopwords, apply stemming or lemmatization, and create features like word count and sentiment score.

Code Screenshot :

```
243 import pandas as pd
244 import re
245 import nltk
246 from nltk.corpus import stopwords
247 from nltk.tokenize import word_tokenize
248 from nltk.stem import PorterStemmer, WordNetLemmatizer
249 from textblob import TextBlob
250 # Download required resources (run once)
251 nltk.download('punkt')
252 nltk.download('stopwords')
253 nltk.download('wordnet')
254 # Load dataset
255 # Replace 'social_media.csv' with your file
256 df = pd.read_csv(r"C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19\social_media.csv")
257 print("Original Data:")
258 print(df.head())
259 # -----
260 # 1. Remove Duplicates & Nulls
261 # -----
262 df = df.drop_duplicates()
263 df = df.dropna(subset=['text'])
264 # -----
265 # 2. Clean Text
266 # -----
267 def clean_text(text):
268     text = re.sub(r"http\S+|www\S+", "", text) # remove URLs
269     text = re.sub(r"[^a-zA-Z\s]", "", text) # remove special characters
270     text = text.lower() # convert to lowercase
271     return text
272 df['clean_text'] = df['text'].apply(clean_text)
273 # -----
274 # 3. Tokenization
275 # -----
276 df['tokens'] = df['clean_text'].apply(word_tokenize)
277 # -----
278 # 4. Remove Stopwords
279 # -----
280 stop_words = set(stopwords.words('english'))
281 df['tokens'] = df['tokens'].apply(
282     lambda words: [word for word in words if word not in stop_words]
283 )
284 # -----
285 # 5. Stemming & Lemmatization
286 # -----
287 stemmer = PorterStemmer()
288 lemmatizer = WordNetLemmatizer()
289 # Apply lemmatization (preferred)
290 df['processed_text'] = df['tokens'].apply(
291     lambda words: [lemmatizer.lemmatize(word) for word in words]
292 )
293 # -----
294 # 6. Feature Engineering
295 # -----
296 # Word count
297 df['word_count'] = df['processed_text'].apply(len)
298 # Sentiment score
299 df['sentiment'] = df['clean_text'].apply(
300     lambda text: TextBlob(text).sentiment.polarity
301 )
302 # -----
303 # Final Output
304 # -----
305 print("\nProcessed Data:")
306 print(df[['text', 'processed_text', 'word_count', 'sentiment']].head())
307 # Save cleaned data
308 df.to_csv('processed_social_media.csv', index=False)
309 print("\nText preprocessing completed and saved as 'processed_social_media.csv'")
```

Output Screenshot:

```
PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19> c; cd 'c:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19'; & 'c:\python314\python.exe' 'c:\Users\sakir\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bu
ndled\libs\debugpy\launcher' '52007' '--' 'c:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19\lab19.py'
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\sakir\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt.zip.
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\sakir\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\sakir\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
Original Data:
  post_id  user  text  platform
0        1  user1  I love this product! It's amazing 🍕 http://exa...  Twitter
1        2  user2  Worst service ever!!! Totally disappointed 🤬  Facebook
2        3  user3  Had a great day at the beach #fun #sunshine  Instagram
3        4  user4  Check this out www.link.com awesome stuff!  Twitter
4        5  user5  I'm so tired of this traffic... ugh!!!  Twitter

Processed Data:
  post_id  user  text  processed_text  word_count  sentiment
0        1  user1  I love this product! It's amazing 🍕 http://exa...  [love, product, amazing]  3  0.550
1        2  user2  Worst service ever!!! Totally disappointed 🤬  [worst, service, ever, totally, disappointed]  5  -0.875
2        3  user3  Had a great day at the beach #fun #sunshine  [great, day, beach, fun, sunshine]  5  0.550
3        4  user4  Check this out www.link.com awesome stuff!  [check, awesome, stuff]  3  1.000
4        5  user5  I'm so tired of this traffic... ugh!!!  [im, tired, traffic, ugh]  4  -0.400

Text preprocessing completed and saved as 'processed_social_media.csv'
PS C:\Users\sakir\OneDrive\Desktop\Btech\3 yr 2sem\Ai-Assistant\dataset19> 
```

Explanation/Justification/Observation (100 words / 5 – 6 sentence) :

The social media text dataset was effectively cleaned and preprocessed to improve its quality and usability. Duplicate and null entries were removed, ensuring data consistency. Text was cleaned by eliminating URLs, special characters, and converting to lowercase. Tokenization, stopword removal, and lemmatization enhanced text structure and meaning. Additional features like word count and sentiment score provided deeper insights. Overall, the processed dataset became more structured, meaningful, and suitable for tasks like sentiment analysis, text classification, and natural language processing applications.